Flight Fare Intelligence

What drives domestic flight fares in India, and a dashboard that predicts them



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Project at a glance

445,366

flights, 9 airlines, 42 routes

0.909

R^2 on flights it has not seen

₹3,540

mean absolute error per fare

4 tabs

live dashboard and fare predictor

Explore

12 charts show how class, airline, route, stops, timing and booking window move fares.

Predict

A gradient-boosted model estimates the fare of any trip, within about ₹3,500 on average.

Deploy

A filterable Streamlit app on Streamlit Cloud, with booking tips generated from the data.

Data and cleaning

What is in the data

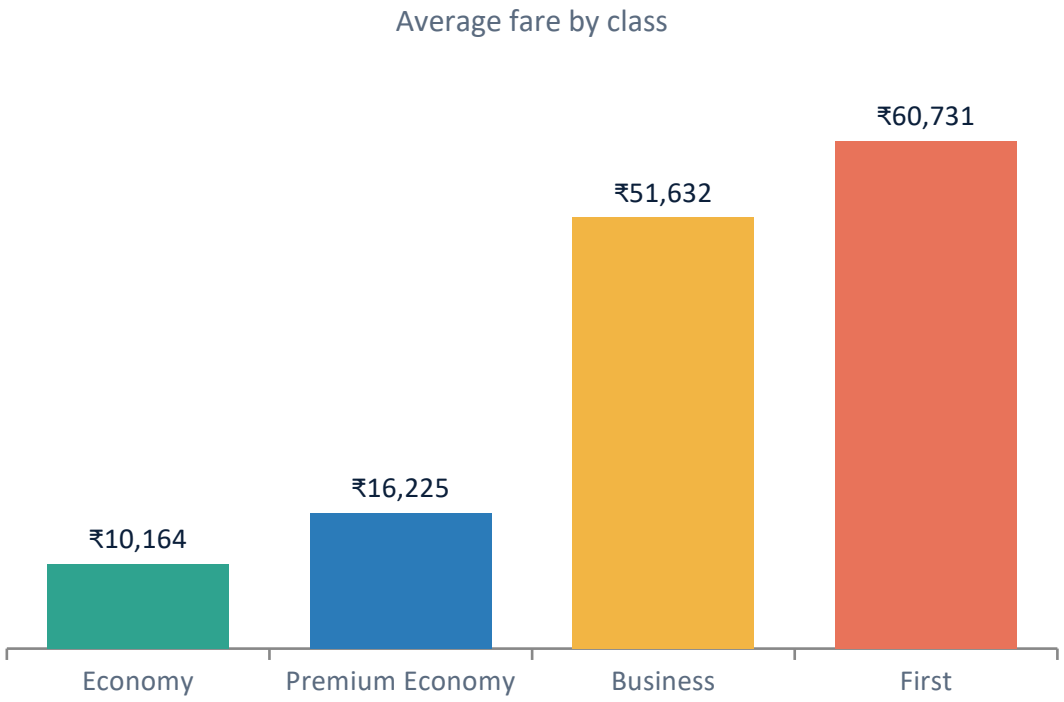
- 445,366 flights after cleaning (452,088 raw rows)
- 7 cities, 9 airlines, 42 routes
- Economy 56%, Business 28%, Premium Economy 16%; First has only 144 rows
- Departures 16 Jan to 6 Mar 2023, 1 to 50 days before takeoff
- Duration is total trip time, including layovers

Cleaning steps

1. Strip spaces and convert numeric columns
2. Drop missing values and 6,722 duplicate rows (1.5%)
3. Drop Date_of_journey and Flight_code
4. Convert stops to 0 / 1 / 2+ and add Route

Limitation: all prices are fare quotes collected on a single day, not bookings. Fare vs days-left patterns are real, but predictions are estimates, not live booking prices.

Class sets the price floor



1.6x Premium Economy, ₹16,225 vs Economy

5.1x Business, ₹51,632 vs Economy

6.0x First, ₹60,731 (only 144 flights) vs Economy

Last-minute flights cost more



Fare within 3 days vs 22+ days ahead

+49% Economy: ₹13,461 vs ₹9,054

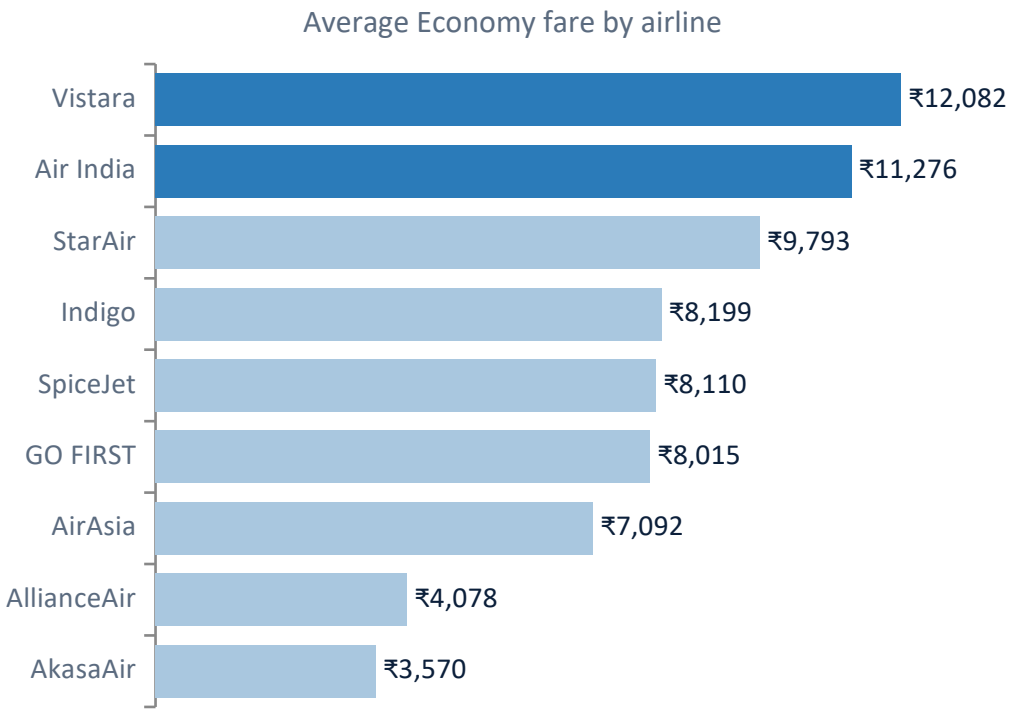
+89% Premium Economy

+22% Business

-1% First: 144 flights, no signal

All prices were collected on one day, so days left is entangled with calendar date. Read it as a pattern, not a law.

Full-service airlines cost more



Premium carriers

Vistara (₹12,082) and Air India (₹11,276) average 38 to 47% more than Indigo (₹8,199).

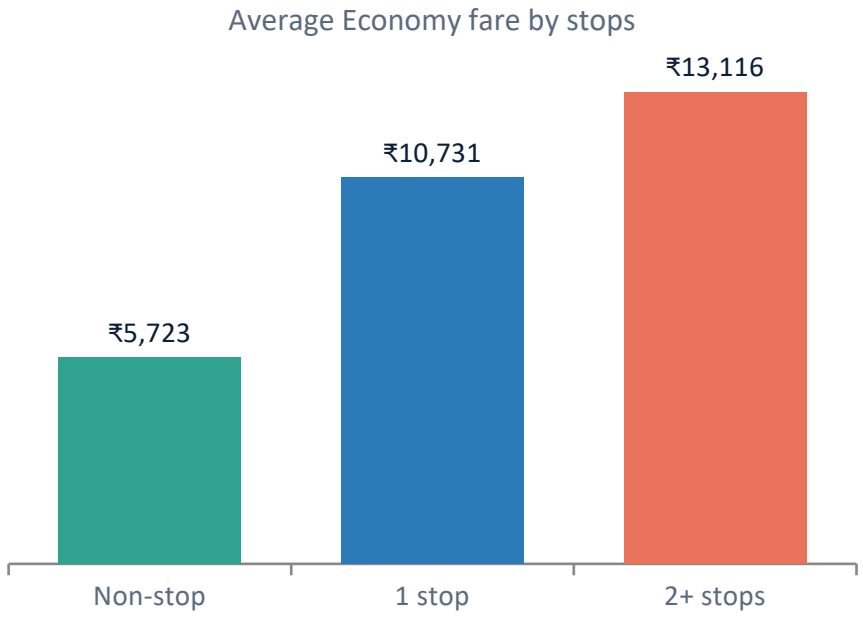
Cheapest and steadiest

AkasaAir averages ₹3,570, partly because it flies short trips (1.6 h vs 5.3 h for Indigo). AllianceAir has the steadiest fares: 31% spread vs 57% for Indigo.

Class and route mix mislead

Only Air India and Vistara sell Business or First, and airlines fly different routes, so averages overstate some airlines.

Stops and timing move fares



Non-stop flights are only 11.6% of all flights.

+88% 1 stop vs non-stop (₹10,731 vs ₹5,723), but about 12.0 h vs 2.1 h door to door, layovers included

+19% 6 AM to 12 PM departures vs before 6 AM (₹10,696 vs ₹9,004)

8% gap between cheapest and costliest weekday (Tuesday ₹9,824, Sunday ₹10,640)

Modelling approach

10 features
7 categorical, plus stops, duration, days left

Encode
OrdinalEncoder, native categorical splits

Boost
HistGradientBoosting on log(1 + fare)

Ship
joblib model, scikit-learn 1.8.0

Model (random split)	MAE (₹)	R^2
Tuned Hist Gradient Boosting	**2,386**	**0.953**
Random Forest	2,357	0.948
Decision Tree	2,587	0.938
Hist Gradient Boosting (untuned)	3,120	0.929
Linear / Ridge regression	4,711	0.862
Mean baseline	16,724	0.000

Why gradient boosting

Highest R^2 and lowest RMSE, MAE within about 1% of Random Forest, roughly 3x faster to train (15 s vs 41 s) and a 5.8 MB file.

Split: 80/20 stratified by class, seed 42 (356,292 train, 89,074 test). Tuning: 3-fold randomized search on 100k training rows. Scores on unseen flights follow on the next slides.

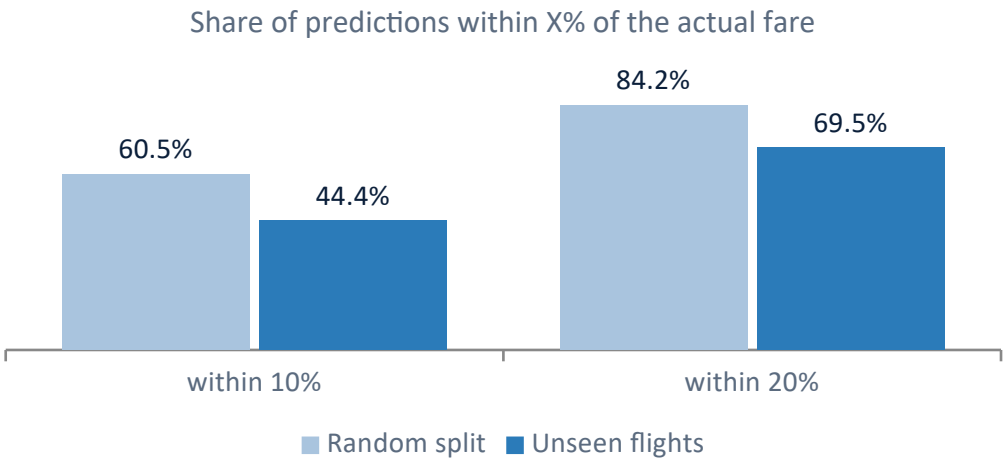
How accurate is the model?

0.909
R^2 on unseen flights

₹3,540
mean absolute error

44.4%
within 10% of actual

69.5%
within 20% of actual



How to read this

- A random split scores higher (R^2 0.953, 60.5% within 10%) because the same flights also appear in training
- Average miss is 14% on Business, 17% on Premium Economy and 18% on Economy; First class is unreliable

Testing on flights the model has not seen

Test	R^2	MAE (₹)	Within 10%	Within 20%	Baseline R^2
Random split	0.953	2,386	60.5%	84.2%	0.922
Unseen flights	**0.909**	**3,540**	**44.4%**	**69.5%**	**0.799**

Why the scores differ

The same flight appears on many dates. In a random split, 99.9% of test rows belong to a flight that is also in training, so part of the score is the model recognising a flight it has seen.

What we report

All rows of a flight stay on one side of the split (5-fold GroupKFold). The model still beats the baseline, the average fare of the same flight (0.909 vs 0.799), so it learns real price patterns.

A split by date was also run, but it is not a true time test: 99.9% of its test flights also appear in training, so it is not reported.

Accuracy by class, with uncertainty

Class	Rows	MAE (₹), 95% CI	Average miss	Within 10%
Economy	248,323	1,814 (1,776 to 1,850)	18.1%	41.8%
Premium Economy	70,364	2,822 (2,728 to 2,918)	17.0%	45.1%
Business	126,535	7,306 (7,098 to 7,534)	14.2%	49.4%
First	144	20,866 (13,860 to 42,072)	31.8%	0.0%

By class

Business is the most reliable (14% average miss), Economy the least (18%). First class has 144 rows from 4 flights and is unreliable.

By airline

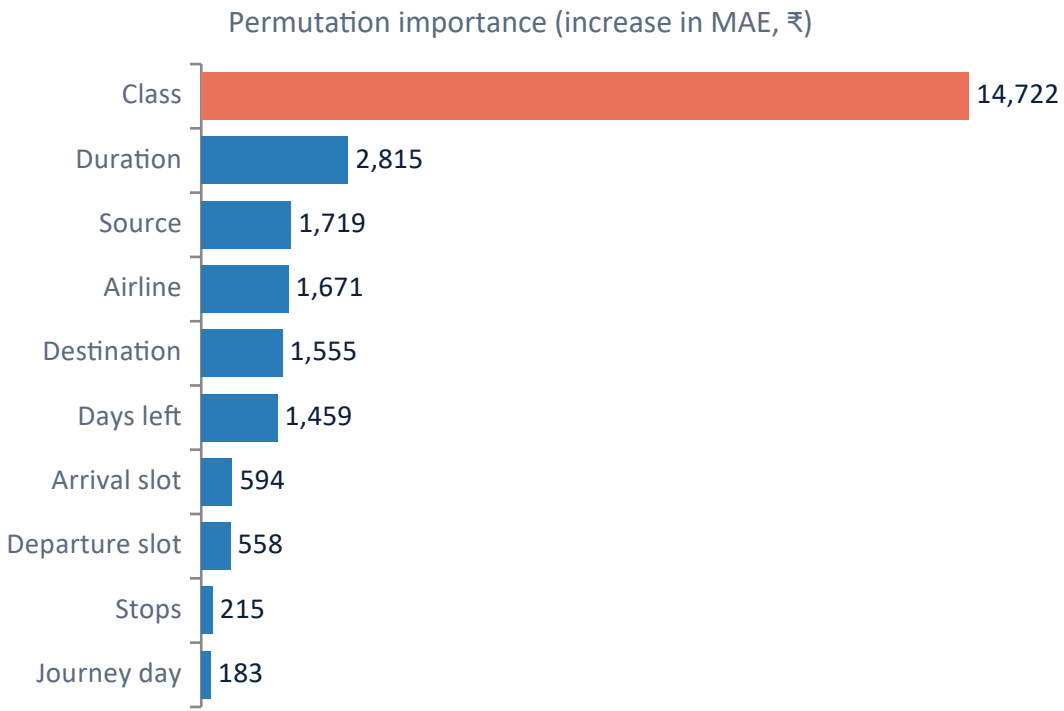
Air India is the most accurate (14.5% average miss), GO FIRST the least (21.5%). Airlines are compared by percentage error because of route mix.

Bias and ranges

No meaningful bias by class, airline, route or days left, except First class, priced about a third too low. An 80% prediction range covered about 76% of unseen flights.

95% intervals resample whole flights, because rows of one flight are not independent. Prediction ranges were tested in the notebook and are not in the app yet.

What drives the fare



Class dominates

Shuffling it adds ₹14,722 of error, 5x the next feature.

Route counts twice

Source (₹1,719) and destination (₹1,555) together rival duration.

Booking window is mid-table

Days left (₹1,459) is on par with airline (₹1,671).

Timing is minor

Departure, arrival, stops and weekday are each under ₹600.

The dashboard

Overview

Class premium, days-left curve, fare distribution

Airlines & Routes

Airline gap, spread, route heatmap, top 10, class mix

Timing, Stops & Duration

Time of day, weekday, stops, duration vs fare

Fare Predictor

Trip inputs to fare, typical error and airline comparison



12 analysis charts and 2 predictor charts, one per row, mobile-friendly.

The fare predictor

- Eight inputs: class, route, airline, stops, days left, times, weekday, duration
- Duration auto-fills with the typical value for that airline, class, route and stops
- Shows the predicted fare against the route median and the middle 50% of route fares
- Shows the typical miss for the chosen class; First class carries a warning
- Plots the fare across 1 to 50 days and the same trip on every airline
- Warns when the combination is not in the data



Filters change every number

All classes



Class filter: Economy only



₹22,920 to ₹10,164

average fare

445,366 to 248,323

flights

+26% to +49%

last-minute premium

Smart Booking Tips are the exception: they use the full dataset and stay fixed.

Engineering challenges solved

Memory limits

Copying 445k rows on every click ran out of memory. Fix: one cached copy, a 12-column filtered frame, a 50-row pre-binned histogram and sampled box and scatter plots.

Model version mismatch

The saved model failed with a scikit-learn _RemainderColsList error. Fix: re-saved it in Colab on the same version as the app, 1.8.0.

Deployment error

Streamlit Cloud raised No module named '_loss'. Fix: pinned all seven packages in requirements.txt.

Debug visibility

The sidebar shows whether the saved or the fallback model is serving. If loading fails, a matching pipeline trains itself.

Limitations and next steps

Limitations

- One-day fare quotes, not bookings: no price-over-time signal
- Only about 7 weeks of departures, no festivals or holidays
- Duration includes layovers, so it reflects connection type
- First class: 144 flights, priced about a third too low
- Random-split scores overstate accuracy; use the unseen-flight numbers

Next steps

- Collect fares across many days and peak periods
- Add holidays, fuel prices and seat availability
- Show prediction ranges in the app
- Time-based validation once multi-day data exists
- MLflow, DVC and a CI smoke test

Try it yourself

Live dashboard and full code, free to explore



App: flight-fare-intelligence.streamlit.app

Code: github.com/Pratham719/flight-fare-intelligence

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